# Point-slope and slope-intercept forms of a line

In this lesson we'll look at two options for writing the equation of a line: point-slope form and slope-intercept form.

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|---|---|
| Point-slope form | $y - y_1 = m(x - x_1)$ |
| Slope-intercept form | $y = mx + b$ |

Both forms require that we know at least two of the following pieces of information about the line:

1. One point, $(x_1, y_1)$
2. A second point, $(x_2, y_2)$
3. The slope, $m$
4. The $y$-intercept, $b$ (the $y$-value where the line crosses the $y$-axis)

If we know any two of these values, we can find the equation of the line in both forms.

## Point-slope form

The equation of a line in point-slope form is

$$y - y_1 = m(x - x_1)$$

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In this form, $(x_1, y_1)$ is a point on the line, and $m$ is the slope. To use this form when we know two points on the line but we don't know the slope, we'll find $m$ as

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Then we'll plug the slope $m$ and the point $(x_1, y_1)$ into point-slope form. Let's do an example where we know the slope and one point on the line.

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**Example**

Write the equation of the line in point-slope form.

$$m = -\frac{1}{4}$$

$$(-6,1)$$

Since we've been given the slope of the line and a point on the line, we can use the point-slope form to find the equation of the line. We'll plug $m = -1/4$ and the point $(-6,1)$ into the point-slope form of the equation of a line.

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{1}{4}(x - (-6))$$

$$y - 1 = -\frac{1}{4}(x + 6)$$

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Let's try an example where we know two points on the line.

**Example**

Find the point-slope form of the equation of the line that passes through the points $(-2, -4)$ and $(3,5)$.

We'll start by finding the slope of the line. It never matters which point we use for $(x_1, y_1)$ and which one we use for $(x_2, y_2)$, as long as we stay consistent. Let's set

$$(x_1, y_1) = (-2, -4)$$

$$(x_2, y_2) = (3,5)$$

Plug these into the formula for slope.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{5 - (-4)}{3 - (-2)}$$

$$m = \frac{9}{5}$$

Next, substitute $m = 9/5$ and the point $(-2, -4)$ into point-slope form.

$$y - (-4) = \frac{9}{5}(x - (-2))$$

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$$y + 4 = \frac{9}{5}(x + 2)$$

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## Slope-intercept form

The equation of a line in slope-intercept form is

$$y = mx + b$$

where $m$ is the slope and $b$ is the $y$-intercept. Since the $y$-intercept, $b$, is the $y$-coordinate of the point at which the graph crosses the $y$-axis, and since the $x$-coordinate of every point on the $y$-axis is $0$, the point where the graph crosses the $y$-axis is $(0,b)$.

Let's try an example where we know the slope and a point on the line.

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**Example**

Find the equation of the line in slope-intercept form.

$$m = -3$$

$$(0,1)$$

If we recognize that the point $(0,1)$ lies on the $y$-axis (since its $x$-coordinate is $0$), then we can recognize that the $y$-intercept is $y = b = 1$, and we can substitute $m = -3$ and $b = 1$ into slope-intercept form.

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$$y = mx + b$$

$$y = -3x + 1$$

Even if we were unsure about whether $(0,1)$ gave us the $y$-intercept, we could still plug the slope $m = -3$ and the point $(0,1)$ into the point-slope form of the equation of a line.

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -3(x - 0)$$

$$y - 1 = -3x$$

$$y = -3x + 1$$

Let's try an example where we know two points on the line.

**Example**

Find the slope-intercept form of the equation of the line that passes through $(-1, -2)$ and $(3, -4)$.

First we need to find the slope, so let's set

$$(x_1, y_1) = (-1, -2)$$

$$(x_2, y_2) = (3, -4)$$

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and then plug these points into the slope formula.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-4 - (-2)}{3 - (-1)}$$

$$m = \frac{-2}{4}$$

$$m = -\frac{1}{2}$$

Next, substitute $m = -1/2$ and $(x_1, y_1) = (-1, -2)$ into the point-slope form of the equation of a line.

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = -\frac{1}{2}(x - (-1))$$

$$y + 2 = -\frac{1}{2}(x + 1)$$

$$y + 2 = -\frac{1}{2}x - \frac{1}{2}$$

$$y = -\frac{1}{2}x - \frac{5}{2}$$

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## Converting between the forms

Regardless of which form we start with, point-slope or slope-intercept, we can easily convert back and forth between the two.

We've worked four examples so far in this lesson. In this first two, we found equations in point-slope form,

$$y-1=-\frac{1}{4}(x+6) \qquad y+4=\frac{9}{5}(x+2)$$

To convert equations in point-slope form to equations in slope-intercept form, we simplify the right side of the equation and then solve for $y$.

$$y-1=-\frac{1}{4}x-\frac{6}{4} \qquad y+4=\frac{9}{5}x+\frac{18}{5}$$

$$y=-\frac{1}{4}x-\frac{6}{4}+1 \qquad y=\frac{9}{5}x+\frac{18}{5}-4$$

$$y=-\frac{1}{4}x-\frac{6}{4}+\frac{4}{4} \qquad y=\frac{9}{5}x+\frac{18}{5}-\frac{20}{5}$$

$$y=-\frac{1}{4}x-\frac{1}{2} \qquad y=\frac{9}{5}x-\frac{2}{5}$$

Now the equations are in slope-intercept form, $y=mx+b$.

In the third and fourth examples, we found equations in slope-intercept form,

$$y=-3x+1 \qquad y=-\frac{1}{2}x-\frac{5}{2}$$

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To convert equations in slope-intercept form to equations in point-slope form, we factor out the coefficient on $x$,

$$y = -3\left(x - \frac{1}{3}\right) \qquad\qquad y = -\frac{1}{2}(x + 5)$$

Now the equations are in slope-intercept form, $y - y_1 = m(x - x_1)$. When we convert this way, we'll always have $y_1 = 0$.

## When lines can't be written in either form

Perfectly vertical lines can't be written in either point-slope or slope-intercept forms, because the slope of a vertical line is undefined.

Given two points on a vertical line $(x_1, y_1)$ and $(x_2, y_2)$, we know that $x_1 = x_2$, which means $x_2 - x_1 = 0$ and the slope of the line is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 - y_1}{0}$$

which is undefined because division by $0$ is undefined. So instead of writing vertical lines in point-slope or slope-intercept form, we'll write them as $x = c$. For example, $x = 3$ is the equation of the vertical line that passes through the point $(3,0)$ on the horizontal axis.

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